

Name:

Period:

## Semester 2 Final Review

## Chapter 6A/6B: Rational and Radical Functions

Simplify each expression (rationalize all denominators if applicable):

|                                                                                                           |                                                                                                                                                                                                                                                                      |                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1) $\sqrt{80x^2}$<br>$= \sqrt{16 \cdot 5 \cdot x^2}$<br>$= 4x\sqrt{5}$                                    | 2) $\sqrt{1000a^3c^2}$<br>$= \sqrt{100 \cdot 10 \cdot a^2 \cdot a \cdot c^2}$<br>$= 10ac\sqrt{10a}$                                                                                                                                                                  | 3) $\sqrt{\frac{360x^4}{12}} = \sqrt{30 \cdot (x^2)^2}$<br>$= x^2\sqrt{30}$                                                                                        |
| 4) $-\sqrt[3]{-27y}$<br>$= -1 \cdot \sqrt[3]{(-3)^3 \cdot y}$<br>$= 3\sqrt[3]{y}$                         | 5) $\sqrt{32x^4yz^7}$<br>$= \sqrt{16 \cdot 2 \cdot (x^2)^2 \cdot y \cdot (z^3)^2 \cdot z}$<br>$= 4x^2z^3\sqrt{2yz}$                                                                                                                                                  | 6) $\sqrt{8x^3y} \cdot \sqrt{3x^5y}$<br>$= \sqrt{24x^8y^2}$<br>$= \sqrt{4 \cdot 6 \cdot (x^4)^2 \cdot y^2}$<br>$= 2x^4y\sqrt{6}$                                   |
| 7) $\frac{3}{\sqrt[3]{2x}} \cdot \frac{\sqrt[3]{4x^2}}{\sqrt[3]{4x^2}}$<br>$= \frac{3\sqrt[3]{4x^2}}{2x}$ | 8) $-5\sqrt{162} - 5\sqrt{32}$<br>$= -5 \cdot \sqrt{81 \cdot 2} - 5 \cdot \sqrt{16 \cdot 2}$<br>$= -5 \cdot 9\sqrt{2} - 5 \cdot 4\sqrt{2}$<br>$= -45\sqrt{2} - 20\sqrt{2}$<br>$= -65\sqrt{2}$                                                                        | 9) $3\sqrt{x} + 2\sqrt{4x} - 7\sqrt{xy}$<br>$= 3\sqrt{x} + 2 \cdot 2\sqrt{x} - 7\sqrt{xy}$<br>$= 3\sqrt{x} + 4\sqrt{x} - 7\sqrt{xy}$<br>$= 7\sqrt{x} - 7\sqrt{xy}$ |
| 10) $(3 - \sqrt{5})(3 + \sqrt{5})$<br>$= 9 + 3\sqrt{5} - 3\sqrt{5} - 5$<br>$= 4$                          | 11) $\frac{3+\sqrt{2}}{\sqrt{5}-\sqrt{10}} \cdot \frac{(\sqrt{5}+\sqrt{10})}{(\sqrt{5}+\sqrt{10})}$<br>$= \frac{3\sqrt{5} + 3\sqrt{10} + \sqrt{10} + \sqrt{20}}{5-10}$<br>$= \frac{3\sqrt{5} + 4\sqrt{10} + 2\sqrt{5}}{-5}$<br>$= \frac{5\sqrt{5} + 4\sqrt{10}}{-5}$ | 12) $\frac{3}{2-\sqrt{x}} \cdot \frac{(2+\sqrt{x})}{(2+\sqrt{x})}$<br>$= \frac{6 + 3\sqrt{x}}{4 - x}$                                                              |

|                                                                                                                       |                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 13) Rewrite the equation in simplest radical form:<br>$x^{\frac{1}{2}}y^{\frac{2}{3}} = \sqrt{x} \cdot \sqrt[3]{y^2}$ | 14) Rewrite the equation in exponential form:<br>$\sqrt[5]{x^7} = x^{\frac{7}{5}}$ |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|

Find the domain of each equation:

|                                                                                                   |                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 15) $f(x) = \sqrt{7x + 42}$<br>$7x + 42 \geq 0$<br>$7x \geq -42$<br>$x \geq -6$<br>$[-6, \infty)$ | 16) Find the domain of $g(x) = \frac{x-9}{2x+1}$<br>$2x+1 \neq 0$<br>$x \neq -\frac{1}{2}$ $(-\infty, -\frac{1}{2}) \cup (-\frac{1}{2}, \infty)$ |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|

For the functions  $f(x) = x^2 - x - 12$  and  $g(x) = x + 5$  find each of the following:

|                                                                                                                                   |                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 17) $(f + g)(x) = f(x) + g(x)$<br>$= (x^2 - x - 12) + (x + 5)$<br>$= x^2 - 7$                                                     | 18) $(f - g)(x) = f(x) - g(x)$<br>$= (x^2 - x - 12) - (x + 5)$<br>$= x^2 - x - 12 - x - 5$<br>$= x^2 - 2x - 17$ |
| 19) $(fg)(x) = f(x) \cdot g(x)$<br>$= (x^2 - x - 12)(x + 5)$<br>$= x^3 + 5x^2 - x^2 - 5x - 12x - 60$<br>$= x^3 + 4x^2 - 17x - 60$ | 20) $(f \div g)(x) = \frac{f(x)}{g(x)}$<br>$= \frac{x^2 - x - 12}{x + 5} = \frac{(x + 3)(x - 4)}{(x + 5)}$      |
| 21) $(f \circ g)(x) = f(g(x))$<br>$= (x + 5)^2 - (x + 5) - 12$<br>$= x^2 + 10x + 25 - x - 5 - 12$<br>$= x^2 + 9x + 8$             | 22) $(f \circ g)(2) = f(g(2))$<br>$g(2) = 2 + 5 = 7$<br>$f(7) = (7)^2 - (7) - 12$<br>$= 49 - 19$<br>$= 30$      |

Find the inverse of each function:

|                                                                                                                                                   |                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 23) $y = \sqrt[3]{4x + 2}$<br>$x = \sqrt[3]{4y} + 2$<br>$(x - 2)^3 = \sqrt[3]{4y}^3$<br>$(x - 2)^3 = 4y$<br>$\frac{(x - 2)^3}{4} = y = f^{-1}(x)$ | 24) $y = (x + 2)^5 - 3$<br>$x = (y + 3)^5 - 3$<br>$(x + 3) = (y + 3)^5$<br>$\sqrt[5]{x + 3} = y + 3$<br>$\sqrt[5]{x + 3} - 3 = y = f^{-1}(x)$ |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

Solve each equation:

25)  $\sqrt{3x + 2} = 6$

$$3x + 2 = 36$$

$$3x = 34$$

$$x = \frac{34}{3}$$

$$\sqrt{3 \cdot \frac{34}{3} + 2} = 6$$

$$\sqrt{36} = 6$$
  
✓

26)  $\frac{3x^{\frac{2}{3}}}{3} = \frac{12}{3}$

$$3\sqrt[3]{x^2} = 4$$

$$x^2 = 64$$

$$x = \pm 8$$

27)  $\sqrt{x + 5} = (x + 3)^2$

$$x + 5 = x^2 + 6x + 9$$

$$0 = x^2 + 5x + 4$$

$$0 = (x + 4)(x + 1)$$

$$x \neq -4$$

extraneous

$$x = -1$$

$$\sqrt{4} = -1 + 3$$
  
✓

$$\sqrt{1} \neq -4 + 3$$

28)  $3\sqrt[5]{x} - 1 = -7$

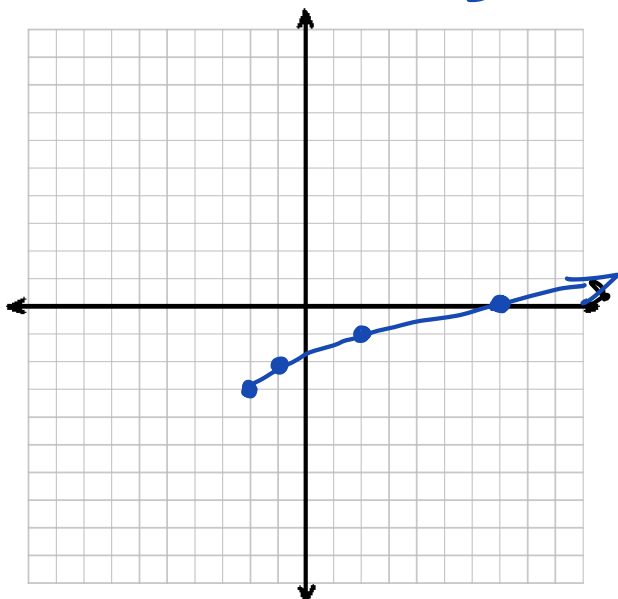
$$3\sqrt[5]{x} = -6$$

$$\sqrt[5]{x} = (-2)^5$$

$$x = -32$$

Plot a minimum of 3 points for each function below.

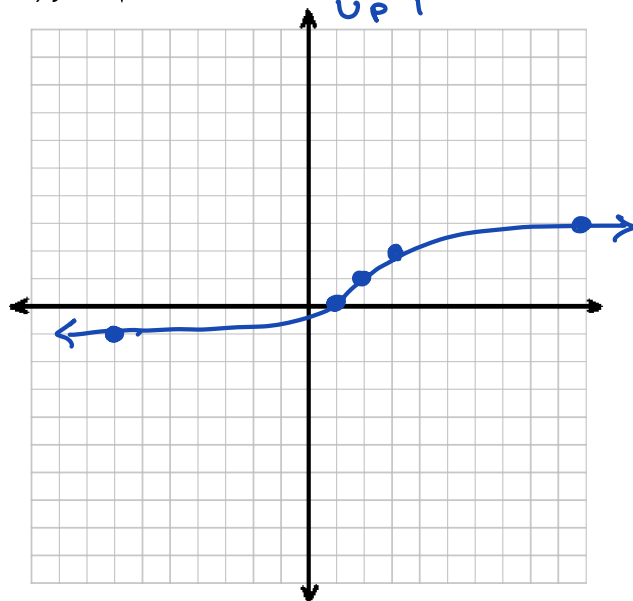
29)  $y = \sqrt{x + 2} - 3$

Left 2  
Down 3

Domain:  $[-2, \infty)$

Range:  $[-3, \infty)$

30)  $y = \sqrt[3]{x - 2} + 1$

Right 2  
Up 1

Domain:  $(-\infty, \infty)$

Range:  $(-\infty, \infty)$

## Chapter 7: Exponential and Logarithmic Functions

Simplify:

|                          |                            |                          |                                     |                                  |                     |
|--------------------------|----------------------------|--------------------------|-------------------------------------|----------------------------------|---------------------|
| 31) $\log 1000$<br>$= 3$ | 32) $\ln e^{32}$<br>$= 32$ | 33) $\log_9 81$<br>$= 2$ | 34) $\log_5 \frac{1}{25}$<br>$= -2$ | 35) $\log 0$<br><i>undefined</i> | 36) $\log 1$<br>$0$ |
|--------------------------|----------------------------|--------------------------|-------------------------------------|----------------------------------|---------------------|

Solve each equation. Round to three decimal places.

|                                                                                                   |                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 37) $9^{x+1} = 27^{x+1}$<br>$3^{2(x+1)} = 3^{3(x+1)}$<br>$2x+2 = 3x+3$<br>$-1 = x$                | 38) $7^x = 98$<br>$\log_7(98) = x$<br>$x \approx 2.356$                                                                                                                                                                                             |
| 39) $\log_3(2x-1) = 2$<br>$3^2 = 2x-1$<br>$9 = 2x-1$<br>$10 = 2x$<br>$5 = x$<br>$\log_3(9) = 2$ ✓ | 40) $\log_2(x+4) + \log_2(x) = 5$<br>$\log_2(x^2+4x) = 5$<br>$2^5 = x^2+4x$<br>$0 = x^2+4x-32$<br>$0 = (x+8)(x-4)$<br>$x \neq -8$ <span style="border: 1px solid black; padding: 2px;"><math>x = 4</math></span><br>$\uparrow$<br><i>extraneous</i> |

Expand:

|                                                           |                                                                                    |
|-----------------------------------------------------------|------------------------------------------------------------------------------------|
| 41) $\log x^2 y^3$<br>$2 \cdot \log(x) + 3 \cdot \log(y)$ | 42) $\log\left(\frac{xy}{5z^2}\right)$<br>$\log(x) + \log(y) - \log(5) - 2\log(z)$ |
|-----------------------------------------------------------|------------------------------------------------------------------------------------|

Condense:

|                                                                    |                                                                                                                           |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 43) $3\log(3) + 7\log(2)$<br>$\log(3^3 \cdot 2^7)$<br>$\log(3456)$ | 44) $4\log x + 2\log y - z + \log y$<br>$\log\left(\frac{x^4 y^2 \cdot y}{z}\right) = \log\left(\frac{x^4 y^3}{z}\right)$ |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|

45) The current student enrollment of Paly is 2000 students and expected to decline 2% each year for the next ten years. If these projections are correct, what will Paly's enrollment be in ten years? Create a mathematical model and round to the nearest student.

$$y = a(1 \pm r)^t$$

$$y = 2000(1 - 0.02)^{10} \approx 1634 \text{ students}$$

46) Professor Rowe deposits \$7800 into a savings account that pays 9.2% annual interest compounded yearly. How long will it take for his account to have \$20,000? Write the equation required and use algebra to solve. Provide your answer in years with three decimal places.

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

$$\frac{20000}{7800} = \frac{7800}{7800} \left(1 + \frac{0.092}{1}\right)^{1(t)}$$

$$t = \log_{(1.092)} \left(\frac{100}{39}\right)$$

$$t \approx 10.699 \text{ years}$$

$$\frac{100}{39} = (1.092)^t$$

47) You put \$15,000 in a money market account that pays 2.3% compounded continuously. How much do you have after 12 years?

$$A = Pe^{rt} = 15000e^{(0.023)(12)}$$

$$\approx \$19767.72$$

48) You put \$40,000 on a credit card with 19% interest compounded continuously. If you don't pay off any of your debt, how long will it take to reach \$100,000? (This credit card has no other fees or penalties)

$$A = Pe^{rt}$$

$$\frac{100000}{40000} = \frac{40000}{40000} e^{0.19t}$$

$$\frac{5}{2} = e^{0.19t}$$

$$\ln\left(\frac{5}{2}\right) = 0.19t$$

$$t = \frac{\ln\left(\frac{5}{2}\right)}{0.19}$$

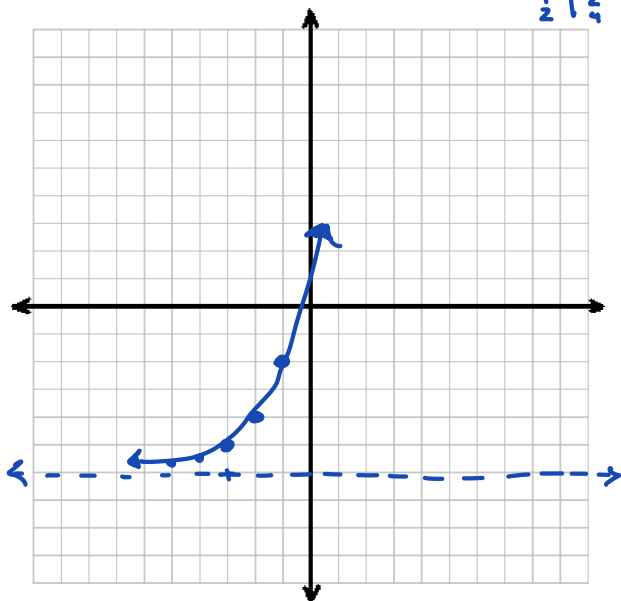
$$t \approx 4.823 \text{ years}$$

Plot a minimum of 3 points for each function below.

49)  $f(x) = (2)^{x+3} - 6$

Left 3  
Down 6

$$\begin{array}{r|l} x & y \\ \hline 2 & 2 \\ 1 & 2 \\ 0 & 2 \\ -1 & 2 \\ -2 & 2 \\ -3 & 2 \end{array}$$

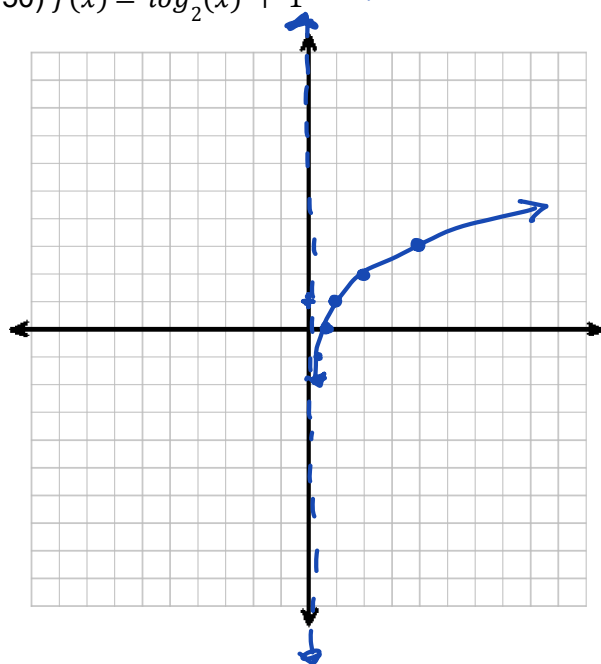


Domain:  $(-\infty, \infty)$

Range:  $(-6, \infty)$

50)  $f(x) = \log_2(x) + 1$

Up 1



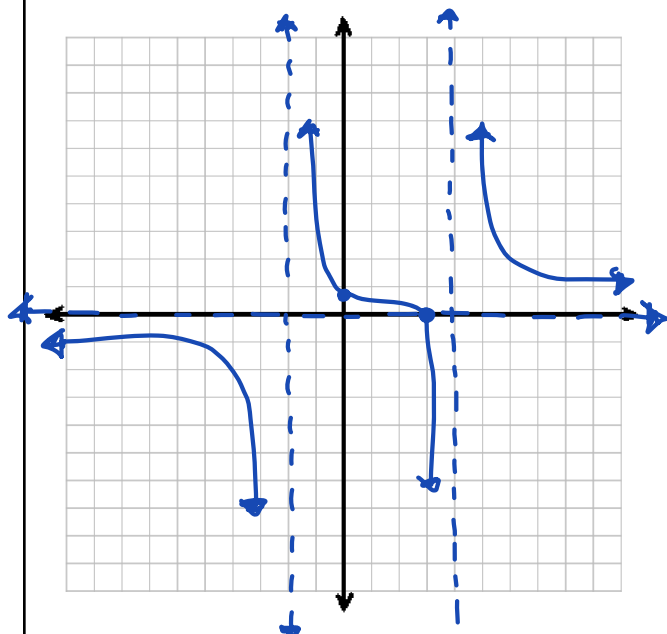
Domain:  $(0, \infty)$

Range:  $(-\infty, \infty)$

## Chapter 8: Rational Functions

Graph each function. Identify the key features.

51)  $y = \frac{(x-3)}{(x+2)(x-4)}$



Horizontal Asymptote:

$y = 0$

num > denom  
Crosses at  
 $0 = \frac{(x-3)}{(x+2)(x-4)}$   
 $x = 3$

Vertical Asymptote:

$x = -2$

$x = 4$

x-intercept(s):

$(3, 0)$

y-intercept(s):

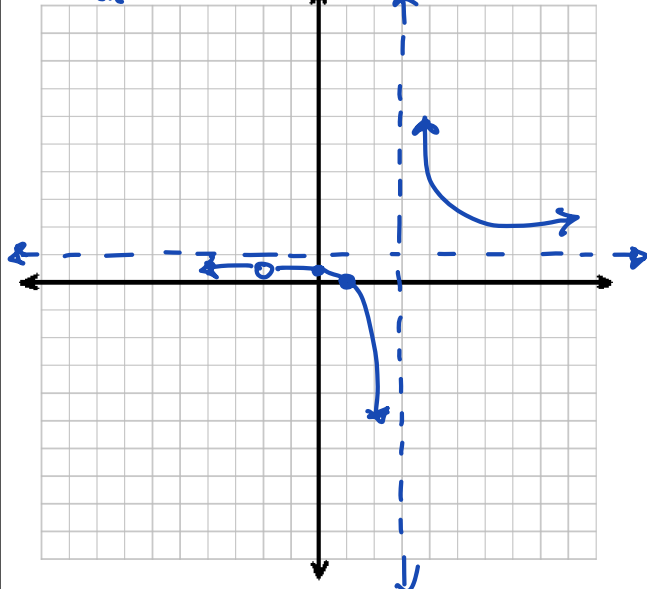
$$y = \frac{(0-3)}{(0+2)(0-4)} = \frac{-3}{-8} \quad \left(0, \frac{3}{8}\right)$$

Hole(s):

None

52)  $y = \frac{(x+2)(x-1)}{(x+2)(x-3)} = \frac{(x-1)}{(x-3)}$

Hole at



Horizontal Asymptote: num = denom

$y = 1$

Vertical Asymptote:

$x = 3$

x-intercept(s):

$$0 = \frac{x-1}{x-3} \quad x = 1 \quad (1, 0)$$

y-intercept(s):

$$y = \frac{0-1}{0-3} = \frac{1}{3} \quad \left(0, \frac{1}{3}\right)$$

Hole(s):

$$y = \frac{-2-1}{-2-3} = \frac{-3}{-5} \quad \left(-2, \frac{3}{5}\right)$$

Simplify.

|                                                                                                                                                                    |                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $53) \frac{2x^4y}{x^3y^2} \cdot \frac{xy^{-3}}{4x^6} = \frac{2x^5y}{4x^9y^2 \cdot y^3}$ $= \frac{1}{2x^4y^4}$                                                      | $54) \frac{3x-x^2}{x^2-2x-3} = \frac{-x(x-3)}{(x-3)(x+1)}$ $= \frac{-x}{x+1}$                                                                                                                                   |
| $55) \frac{1}{x^2+8} \div \frac{1}{x+2}$ $= \frac{1}{(x^2+8)} \cdot \frac{(x+2)}{1}$ $= \frac{x+2}{x^2+8}$                                                         | $56) \frac{x^2-9}{x+1} \div \frac{x-3}{x^2-x-2}$ $= \frac{(x+3)(x-3)}{(x+1)} \cdot \frac{(x-2)(x+1)}{(x-3)}$ $= (x+3)(x-2)$ $\text{or } = x^2 + x - 6$                                                          |
| $57) \frac{x+2}{x-1} + \frac{5}{x+3}$ $= \frac{(x+2)(x+3) + 5(x-1)}{(x-1)(x+3)}$ $= \frac{x^2 + 5x + 6 + 5x - 5}{(x-1)(x+3)}$ $= \frac{x^2 + 10x + 1}{(x-1)(x+3)}$ | $58) \frac{2x-3}{x^2-5x} - \frac{x+4}{x-5}$ $= \frac{(2x-3)}{x(x-5)} + \frac{-(x+4)}{(x-5)} \cdot \frac{x}{x}$ $= \frac{2x-3-x^2-4x}{x(x-5)}$ $= \frac{-x^2-2x-3}{x(x-5)}$                                      |
| $59) \frac{x}{x-2} - \frac{4}{x^2-x}$ $\frac{x}{x-2} + \frac{4}{x-2}$ $\frac{x+4}{x-2}$                                                                            | $60) \frac{5x}{x^2-x-6} + \frac{4}{x^2+4x+4}$ $= \frac{5x(x+2)}{(x-3)(x+2)(x+2)} + \frac{4(x-3)}{(x+2)(x+2)(x-3)}$ $= \frac{5x^2 + 10x + 4x - 12}{(x-3)(x+2)(x+2)}$ $= \frac{5x^2 + 14x - 12}{(x-3)(x+2)(x+2)}$ |



## Chapter 11: Probability

61) If there are 30 students in a class, how many different ways can you make a group of four?

$${}_{30}C_4 = 27405 \text{ ways}$$

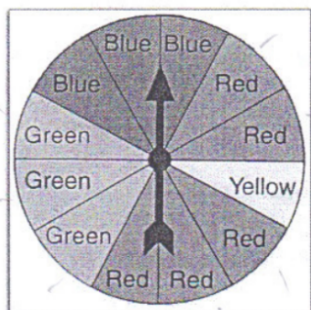
62) First, second, and third place in a spelling bee win an award. If there are 20 contestants, how many ways can the awards be given out?

$${}_{20}P_3 = 6840 \text{ ways}$$

63) What is the total number of different 7-letter arrangements that can be formed using the letters in the word ALGEBRA?

$$\frac{7!}{2!} = 2520 \text{ ways}$$

64) You spin the spinner below once.



a) What is  $P(\text{blue})$ ?  $\frac{3}{12} = \frac{1}{4}$

b) What is  $P(\text{yellow})$ ?  $\frac{1}{12}$

c) What is  $P(\text{red or green})$ ?  $\frac{8}{12} = \frac{2}{3}$

65) The Hiking Club plans to go camping in a State park where the probability of rain on any given day is 0.3. What is the probability that it will rain on exactly two of the five days they are there? Round your answer to the nearest thousandth.

$${}_5C_2 (0.3)^2 (0.7)^3 = 0.309$$

66) If the probability that the Giants will beat the Dodgers in a game is 0.63, what is the probability that the Giants will win at most two out of five games in a series against the Dodgers? Round your answer to the nearest thousandth.

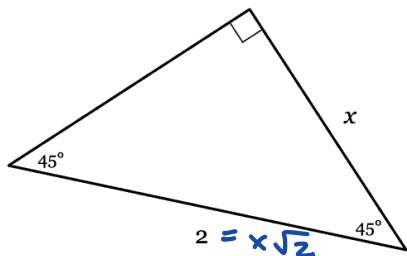
$${}_5C_0 (0.63)^0 (0.37)^5 + {}_5C_1 (0.63)^1 (0.37)^4 + {}_5C_2 (0.63)^2 (0.37)^3$$

$$= 0.267$$

## Chapter T: Trigonometry

Find the side length of  $x$  in simplest radical form with a rational denominator.

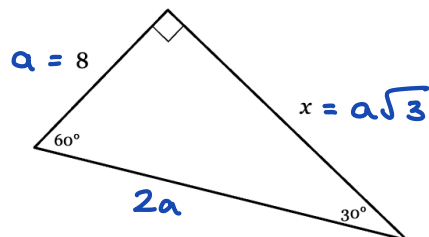
67)



$$x\sqrt{2} = 2$$

$$x = \frac{2}{\sqrt{2}} = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

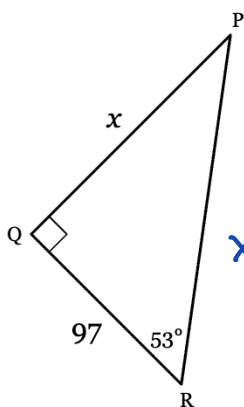
68)



$$x = 8\sqrt{3}$$

Solve for  $x$ . If necessary, round to three decimal places.

69)

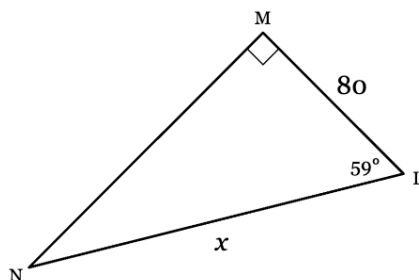


$$\tan(53^\circ) = \frac{x}{97}$$

$$x = 97 \cdot \tan(53^\circ)$$

$$x \approx 128.723$$

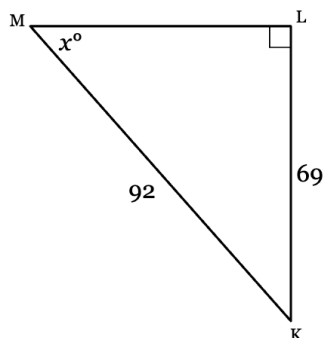
70)



$$\cos(59^\circ) = \frac{80}{x}$$

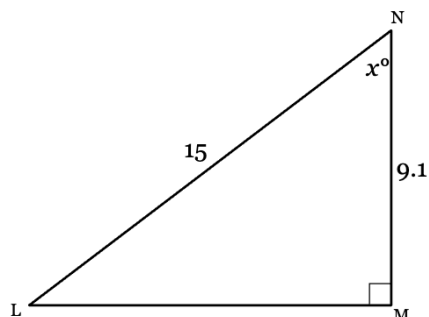
$$x = \frac{80}{\cos(59^\circ)} \approx 155.328$$

71)



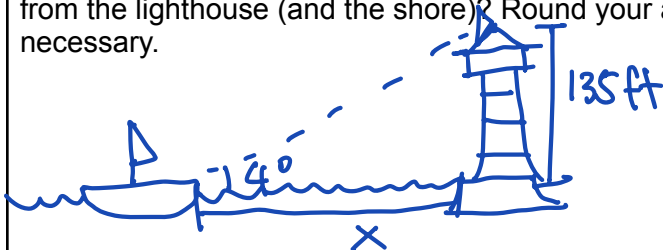
$$x = \sin^{-1}\left(\frac{69}{92}\right) \approx 48.590^\circ$$

72)



$$x = \cos^{-1}\left(\frac{9.1}{15}\right) \approx 52.651^\circ$$

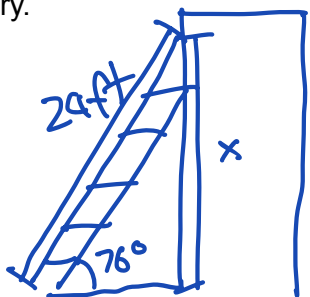
73) A boat is heading towards a lighthouse, whose beacon-light is 135 feet above the water. The boat's crew measures the angle of elevation to the beacon,  $4^\circ$ . What is the ship's horizontal distance from the lighthouse (and the shore)? Round your answer to the nearest hundredth of a foot if necessary.



$$\tan(4^\circ) = \frac{135}{x}$$

$$x = \frac{135}{\tan(4^\circ)} \approx 1930.59 \text{ ft}$$

74) Nora leans a 24-foot ladder against a wall so that it forms an angle of  $76^\circ$  with the ground. How high up the wall does the ladder reach? Round your answer to the nearest hundredth of a foot if necessary.



$$\sin(76^\circ) = \frac{x}{24}$$

$$x = 24 \cdot \sin(76^\circ) \approx 23.29 \text{ ft}$$

75) Convert the following degrees to radians.

$$135^\circ \cdot \frac{\pi}{180} = \frac{3\pi}{4}$$

$$-300^\circ \cdot \frac{\pi}{180} = -\frac{5\pi}{3}$$

$$53^\circ \cdot \frac{\pi}{180} = \frac{53\pi}{180}$$

76) Convert the following radians to degrees.

$$\frac{5\pi}{6} \cdot \frac{180}{\pi} = 150^\circ$$

$$-\frac{3\pi}{4} \cdot \frac{180}{\pi} = -135^\circ$$

$$\frac{7\pi}{3} \cdot \frac{180}{\pi} = 420^\circ$$

77) Use the unit circle to find the sine and cosine of the following angles.

225°

$$\sin(225^\circ) = -\frac{\sqrt{2}}{2}$$

$$\cos(225^\circ) = -\frac{\sqrt{2}}{2}$$

$-450^\circ + 360 + 360 = 270^\circ$

$$\sin(-450^\circ) = 0$$

$$\cos(-450^\circ) = -1$$

$\frac{5\pi}{2}$  radians  $-\frac{4\pi}{2} = \frac{\pi}{2}$

$$\sin\left(\frac{5\pi}{2}\right) = 1$$

$$\cos\left(\frac{5\pi}{2}\right) = 0$$